

SULIT
SK015
Chemistry 1
Semester I
Session 2020/2021
2 hours

SK015
Kimia 1
Semester I
Sesi 2020/2021
2 jam



KEMENTERIAN PENDIDIKAN MALAYSIA

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

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Kertas soalan ini mengandungi **9** halaman bercetak.

*This question paper consists of **9** printed pages.*

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Answer **all** questions.
Jawab semua soalan.

- 1 (a) Sodium metasilicate, Na_2SiO_3 , is used in the production of silica gel.
Natrium metasilikat, Na_2SiO_3 , digunakan dalam pengeluaran gel silika.

(i) Write the isotopic notation for silicon atom.

Tulis notasi isotop untuk atom silikon.

(ii) Calculate the number of Si atoms in 50 g of Na_2SiO_3 .

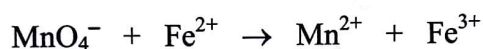
Kirakan bilangan atom Si dalam 50 g Na_2SiO_3 .

[3 marks]

[3 markah]

- (b) The following reaction takes place in an acidic condition.

Tindak balas berikut berlaku dalam keadaan berasid.



Write a balance equation for the above reaction.

Tulis suatu persamaan berimbang untuk tindak balas tersebut.

[3 marks]

[3 markah]

- (c) Compound **X** has a molar mass of 294.20 g. Analysis of **X** shows that it contains 26.58 g of potassium, 35.35 g of chromium and 38.07 g of oxygen. Determine the molecular formula and name **X**.

*Sebatian **X** mempunyai jisim molar 294.20 g. Analisis **X** menunjukkan bahawa ia mengandungi 26.58 g kalium, 35.35 g kromium dan 38.07 g oksigen. Tentukan formula molekul dan namakan **X**.*

[6 marks]

[6 markah]

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- (d) The reaction between nitrogen dioxide, NO_2 , with water produces nitric acid, HNO_3 , and nitrogen monoxide, NO . In a reaction between 100.0 g of NO_2 and 50.0 g of water, 80.0 g of HNO_3 is produced.

Tindak balas antara nitrogen dioksida, NO_2 , dengan air menghasilkan asid nitrik, HNO_3 , dan nitrogen monoksida, NO . Dalam suatu tindak balas antara 100.0 g NO_2 dan 50.0 g air, 80.0 g HNO_3 telah dihasilkan.

- (i) Determine the limiting reactant in the reaction.
Tentukan reaktan penghad di dalam tindak balas.

- (ii) Determine the percentage yield of the reaction.
Tentukan peratusan hasil tindak balas.

[9 marks]

[9 markah]

- 2 (a) Calculate the energy emitted and the wavelength of the spectrum line when an electron moves from $n = 6$ to $n = 3$. State the series and electromagnetic region of the spectrum line.

Kira tenaga yang dipancarkan dan panjang gelombang garis spektrum apabila elektron bergerak dari $n = 6$ ke $n = 3$. Nyatakan siri dan kawasan elektromagnet bagi garis spektrum ini.

[6 marks]

[6 markah]

- (b) An electron has the following quantum numbers: $n = 3$, $\ell = 2$. Identify its orbital. Draw the spatial orientation for all the possible orbitals.

Suatu elektron mempunyai nombor kuantum berikut: $n = 3$, $\ell = 2$. Kenal pasti orbitalnya. Lukis orientasi ruang bagi kesemua orbital yang mungkin.

[4 marks]

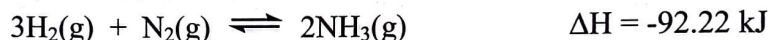
[4 markah]

- 3 Phosgene, COCl_2 , is a chemical used in the production of plastics and pesticides. Given that chlorine cannot be the central atom,
- Fosgena, COCl_2 , adalah bahan kimia yang digunakan dalam pengeluaran plastik dan racun perosak. Diberi klorin tidak boleh menjadi atom pusat,*
- (a) draw **three (3)** possible structures of phosgene.
*lukis **tiga (3)** struktur fosgena yang mungkin.*
- [3 marks]
[3 markah]
- (b) determine the most plausible structure and give your reason.
tentukan struktur yang paling munasabah dan berikan alasan anda.
- [5 marks]
[5 markah]
- (c) determine the hybrid orbital for the central atom of the most plausible structure.
tentukan orbital hibrid atom pusat bagi struktur yang paling munasabah.
- [4 marks]
[4 markah]
- (d) name the molecular shape and draw the overlapping orbitals of COCl_2 .
namakan bentuk molekul dan lukis pertindihan orbital bagi COCl_2 .
- [5 marks]
[5 markah]

- 4 (a) Sketch a labelled phase diagram of carbon dioxide.
Lakarkan suatu gambar rajah fasa karbon dioksida berlabel. [4 marks]
[4 markah]

- (b) 2.0 g of He and 61 g of O₂ were placed in a 5.0-L tank at 25 °C. Determine the partial pressure and the total pressure of the gas mixture.
2.0 g He dan 61 g O₂ diletakkan dalam tangki 5.0-L pada 25 °C. Tentukan tekanan separa dan jumlah tekanan campuran gas tersebut. [5 marks]
[5 markah]

- 5 (a) Ammonia, NH₃, is produced at 500 °C and 200 atm in the presence of Fe₂O₃ catalyst according to the following equation:
Ammonia, NH₃, dihasilkan pada 500 °C dan 200 atm dengan kehadiran mangkin Fe₂O₃ mengikut persamaan berikut:



Suggest **two (2)** changes that can be made to increase the yield of NH₃.
Explain your suggestions.

*Cadangkan **dua (2)** perubahan yang boleh dibuat untuk meningkatkan hasil NH₃. Terangkan cadangan anda.*

[4 marks]

[4 markah]

- (b) Consider the following reaction:
Pertimbangkan tindak balas berikut:



Determine the direction of the reaction if the concentrations of SO₂(g), SO₃(g) and O₂(g) are 0.2 M, 0.15 M and 8.55 x 10⁻³ M, respectively.

Tentukan arah tindak balas jika kepekatan SO₂(g), SO₃(g) dan O₂(g) masing-masing ialah 0.2 M, 0.15 M dan 8.55 x 10⁻³ M.

[5 marks]

[5 markah]

- 6 (a) Determine the pH of 0.5 M ethanoic acid, CH_3COOH , solution.
[Given: $K_a \text{CH}_3\text{COOH} = 1.8 \times 10^{-5}$]
Tentukan pH larutan asid etanoik, CH_3COOH , 0.5 M.
[Diberi: $K_a \text{CH}_3\text{COOH} = 1.8 \times 10^{-5}$]

[5 marks]

[5 markah]

- (b) In an experiment, 25 mL of 0.5 M CH_3COOH was titrated against 0.5 M NaOH solution using phenolphthalein as indicator. The end point is reached when 25 mL of NaOH solution was added.

Dalam suatu eksperimen, 25 mL 0.5 M CH_3COOH dititrat dengan larutan 0.5 M NaOH menggunakan fenolftalein sebagai penunjuk. Titik akhir dicapai apabila 25 mL larutan NaOH ditambah.

- (i) Determine the concentration of the product.
Tentukan kepekatan produk.
- (ii) Determine the pH of the solution at the end point.
Tentukan pH larutan pada titik akhir.

[9 marks]

[9 markah]

END OF QUESTION PAPER
KERTAS SOALAN TAMAT